

USB4 1.0 ENGINEERING CHANGE NOTICE FORM

Title: Source of Inter-Domain TMU Parameters
Applied to: USB4 Specification Version 1.0

Brief description of the functional changes:

An accurate description of the source of the Inter-Domain TMU Parameters for Host Routers and Device Routers
--

Benefits as a result of the changes:

A consistent description that matches both Inter-Domain Link on a Host Router and a Device Router

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
--

None

An analysis of the hardware implications:
--

None

An analysis of the software implications:
--

None

An analysis of the compliance testing implications:
--

None

USB4 1.0 ENGINEERING CHANGE NOTICE FORM

Actual Change

(a). 7.3.2 – Inter-Domain Time Sync

To Text:

~~After calculating the Inter-Domain time offset, Inter-Domain frequency offset and Inter-Domain time stamp, if~~
~~the~~ the Router with the IDTI Port ~~is on a Host Router, it~~ shall update the *InterDomainTimeStamp*,
FreqOffsetFromInterDomainHR and *TimeOffsetFromInterDomainHR* fields in Router Configuration Space. If the
IDTI Port is on a Device Router, it shall prepare an Inter-Domain Time Stamp Packet as described in Section
7.3.3.3 where:

(b). Table 7-8 – Follow-Up Packet Payload

To Text:

DW	Bits	Field	Description
	15:0	<i>IDTimeStamp</i>	For a Host Router: If the <i>IDE</i> bit is set to 1b, this field contains the most recent value of the Inter-Domain time stamp received from the IDTI Port. If the <i>IDE</i> bit is set to 0b, then this field shall be set to 0. For a Device Router: This field shall contain the <i>IDTimeStamp</i> value from the last Follow-Up Packet that the IDTI PortRouter received.
9	31:0		
10	31:0		
11	31:0	<i>TimeOffsetFromInterDomainHR</i>	For a Host Router: If the <i>IDE</i> bit is set to 1b, this field contains value of the <i>TimeOffsetFromInterDomainHR</i> field in the last received Inter-Domain Time Stamp Packet. If the <i>IDE</i> bit is set to 0b, then this field shall be set to 0. For a Device Router: This field shall contain the <i>TimeOffsetFromInterDomainHR</i> value in the last Follow-Up Packet that the IDTI PortRouter received.
12	31:0		
13	31:0	<i>FreqOffsetFromInterDomainHR</i>	For a Host Router: If the <i>IDE</i> bit is set to 1b, this field contains value of the <i>FreqOffsetFromInterDomainHR</i> field in the last received Inter-Domain Time Stamp Packet. If the <i>IDE</i> bit is set to 0b, then this field shall be set to 0. For a Device Router: This field shall contain the <i>FreqOffsetFromInterDomainHR</i> value in the last Follow-Up Packet that the IDTI PortRouter received.

USB4 1.0 ENGINEERING CHANGE NOTICE FORM

(c). Table 8-4 – TMU Router Configuration Capability Fields

To Text:

DW	Register Name	Bit(s)	Field Name and Description	Type	Default Value
16	TMU_RTR_CS_1 6	31:0	Inter-Domain Time Stamp Low For a Host Router n Inter-Domain Time Initiator : This field shall contain the least significant 32 bits of the computed -value of the Inter-Domain time stamp <u>received in the last Inter-Domain Time Stamp Packet.</u> The time stamp shall be computed as described in Section 7.4.2.1. For all other Device Routers: This field shall contain the least significant 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0
17	TMU_RTR_CS_1 7	31:0	Inter-Domain Time Stamp Middle For a Host Router n Inter-Domain Time Initiator : This field shall contain the middle 32 bits of the computed -value of the Inter-Domain time stamp <u>received in the last Inter-Domain Time Stamp Packet.</u> The time stamp shall be computed as described in Section 7.4.2.1. For all other Device Routers: This field shall contain the middle 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0
18	TMU_RTR_CS_1 8	15:0	Inter-Domain Time Stamp High For an Inter-Domain Time Initiator: This field shall contain the most significant 32 bits of the computed -value of the Inter-Domain time stamp <u>received in the last Inter-Domain Time Stamp Packet.</u> The time stamp shall be computed as described in Section 7.4.2.1. For all other Device Routers: This field shall contain the most significant 32 bits of the IDTimeStamp value contained in the last received Follow-Up Packet.	RO	0